

# Objective

The objective of this analysis is to investigate the relationship between Bitcoin market sentiment (Fear & Greed Index) and trader performance using historical trading data from Hyperliquid. By combining both datasets, the analysis aims to identify patterns in profitability, trading behavior, and market sentiment that can provide actionable insights for improving trading strategies.

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Display all columns if needed
pd.set_option('display.max_columns', None)
```

```
In [3]: trader_df = pd.read_csv("historical_data (1).csv")
sentiment_df = pd.read_csv("fear_greed_index (1).csv")
```

```
In [4]: print("Trader Dataset")
display(trader_df.head())

print("\nSentiment Dataset")
display(sentiment_df.head())
```

Trader Dataset

|   | Account                                    | Coin | Execution Price | Size Tokens | Size USD | Side |
|---|--------------------------------------------|------|-----------------|-------------|----------|------|
| 0 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9769          | 986.87      | 7872.16  | BUY  |
| 1 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9800          | 16.00       | 127.68   | BUY  |
| 2 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9855          | 144.09      | 1150.63  | BUY  |
| 3 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9874          | 142.98      | 1142.04  | BUY  |
| 4 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9894          | 8.73        | 69.75    | BUY  |



Sentiment Dataset

|   | timestamp  | value | classification | date       |
|---|------------|-------|----------------|------------|
| 0 | 1517463000 | 30    | Fear           | 2018-02-01 |
| 1 | 1517549400 | 15    | Extreme Fear   | 2018-02-02 |
| 2 | 1517635800 | 40    | Fear           | 2018-02-03 |
| 3 | 1517722200 | 24    | Extreme Fear   | 2018-02-04 |
| 4 | 1517808600 | 11    | Extreme Fear   | 2018-02-05 |

```
In [5]: print("Trader Dataset Shape:", trader_df.shape)
        print("Sentiment Dataset Shape:", sentiment_df.shape)
```

Trader Dataset Shape: (211224, 16)

Sentiment Dataset Shape: (2644, 4)

```
In [6]: print(trader_df.columns)

        print(sentiment_df.columns)
```

Index(['Account', 'Coin', 'Execution Price', 'Size Tokens', 'Size USD', 'Side',  
'Timestamp IST', 'Start Position', 'Direction', 'Closed PnL',  
'Transaction Hash', 'Order ID', 'Crossed', 'Fee', 'Trade ID',  
'Timestamp'],  
 dtype='object')

Index(['timestamp', 'value', 'classification', 'date'], dtype='object')

```
In [7]: trader_df.info()

        sentiment_df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 211224 entries, 0 to 211223
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Account               211224 non-null object
1   Coin                  211224 non-null object
2   Execution Price       211224 non-null float64
3   Size Tokens           211224 non-null float64
4   Size USD              211224 non-null float64
5   Side                  211224 non-null object
6   Timestamp IST         211224 non-null object
7   Start Position        211224 non-null float64
8   Direction             211224 non-null object
9   Closed PnL           211224 non-null float64
10  Transaction Hash      211224 non-null object
11  Order ID              211224 non-null int64
12  Crossed               211224 non-null bool
13  Fee                   211224 non-null float64
14  Trade ID             211224 non-null float64
15  Timestamp             211224 non-null float64
dtypes: bool(1), float64(8), int64(1), object(6)
memory usage: 24.4+ MB
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2644 entries, 0 to 2643
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  -
0   timestamp             2644 non-null  int64
1   value                 2644 non-null  int64
2   classification        2644 non-null  object
3   date                  2644 non-null  object
dtypes: int64(2), object(2)
memory usage: 82.8+ KB

```

```

In [8]: print(trader_df.isnull().sum())

        print(sentiment_df.isnull().sum())

```

```

Account          0
Coin              0
Execution Price   0
Size Tokens       0
Size USD          0
Side              0
Timestamp IST     0
Start Position    0
Direction         0
Closed PnL        0
Transaction Hash  0
Order ID          0
Crossed           0
Fee               0
Trade ID          0
Timestamp         0
dtype: int64
timestamp         0
value             0
classification    0
date              0
dtype: int64

```

```

In [9]: print("Trader duplicates:", trader_df.duplicated().sum())

        print("Sentiment duplicates:", sentiment_df.duplicated().sum())

```

```

Trader duplicates: 0
Sentiment duplicates: 0

```

```

In [10]: trader_df["Timestamp IST"] = pd.to_datetime(
        trader_df["Timestamp IST"],
        dayfirst=True
    )

```

```

In [11]: trader_df["Timestamp IST"].head()

```

```

Out[11]: 0    2024-12-02 22:50:00
        1    2024-12-02 22:50:00
        2    2024-12-02 22:50:00
        3    2024-12-02 22:50:00
        4    2024-12-02 22:50:00
        Name: Timestamp IST, dtype: datetime64[ns]

```

```

In [12]: sentiment_df["date"] = pd.to_datetime(sentiment_df["date"])

```

```

In [13]: sentiment_df["date"].head()

```

```

Out[13]: 0    2018-02-01
        1    2018-02-02
        2    2018-02-03
        3    2018-02-04
        4    2018-02-05
        Name: date, dtype: datetime64[ns]

```

```

In [14]: trader_df["Date"] = trader_df["Timestamp IST"].dt.date
        sentiment_df["Date"] = sentiment_df["date"].dt.date

```

```

In [15]: print("Trader Data:")
        print(trader_df["Date"].min(), "to", trader_df["Date"].max())

```

```
print("\nSentiment Data:")
print(sentiment_df["Date"].min(), "to", sentiment_df["Date"].max())
```

Trader Data:  
2023-05-01 to 2025-05-01

Sentiment Data:  
2018-02-01 to 2025-05-02

```
In [16]: merged_df = pd.merge(
    trader_df,
    sentiment_df[["Date", "classification", "value"]],
    on="Date",
    how="left"
)
```

```
In [17]: print("Merged Dataset Shape:", merged_df.shape)
print("Missing Sentiment Values:", merged_df["classification"].isna().sum())

merged_df.head()
```

Merged Dataset Shape: (211224, 19)  
Missing Sentiment Values: 6

```
Out[17]:
```

|   | Account                                    | Coin | Execution Price | Size Tokens | Size USD | Side |
|---|--------------------------------------------|------|-----------------|-------------|----------|------|
| 0 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9769          | 986.87      | 7872.16  | BUY  |
| 1 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9800          | 16.00       | 127.68   | BUY  |
| 2 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9855          | 144.09      | 1150.63  | BUY  |
| 3 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9874          | 142.98      | 1142.04  | BUY  |
| 4 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9894          | 8.73        | 69.75    | BUY  |

```
In [18]: merged_df = merged_df.dropna(subset=["classification"])
```

```
In [19]: merged_df["classification"].isna().sum()
```

Out[19]: 0

```
In [20]: # Exploratory Data Analysis (EDA)

#This section explores trader behavior and performance under different Bitcoin m
```

```
In [21]: print("Total Trades:", len(merged_df))
print("Unique Traders:", merged_df["Account"].nunique())
print("Coins Traded:", merged_df["Coin"].nunique())
print("Unique Coins:", merged_df["Coin"].unique())
```

Total Trades: 211218

Unique Traders: 32

Coins Traded: 246

Unique Coins: ['@107' 'AAVE' 'DYDX' 'AIXBT' 'GMX' 'EIGEN' 'HYPE' 'SOL' 'SUI' 'DOG  
E'

'ETH' 'kPEPE' 'TRUMP' 'ONDO' 'ENA' 'LINK' 'XRP' 'S' 'BNB' 'BERA' 'WIF'  
'LAYER' 'MKR' 'KAITO' 'IP' 'JUP' 'USUAL' 'ADA' 'BTC' 'PURR/USDC' 'ZRO'  
'@7' '@19' '@21' '@44' '@48' '@11' '@15' '@46' '@61' '@28' '@45' '@9'  
'@41' '@38' 'KSHIB' 'GRASS' 'TAO' 'AVAX' '@2' '@6' '@8' '@10' '@12' '@16'  
'@17' '@35' '@26' '@24' '@32' '@29' '@31' '@33' '@34' '@36' '@37' '@47'  
'@53' '@74' 'RUNE' 'CANTO' 'NTRN' 'BLUR' 'ZETA' 'MINA' 'MANTA' 'RNDR'  
'WLD' 'kBONK' 'ALT' 'INJ' 'STG' 'ZEN' 'MAVIA' 'PIXEL' 'ILV' 'FET' 'STRK'  
'CAKE' 'STX' 'ACE' 'PENDLE' 'AR' 'XAI' 'APE' 'MEME' 'NEAR' 'SEI' 'FTM'  
'MYRO' 'BIGTIME' 'IMX' 'BADGER' 'POLYX' 'OP' 'TNSR' 'MAV' 'TIA' 'MERL'  
'TON' 'PURR' 'ME' 'CRV' 'BRETT' 'CHILLGUY' 'MOODENG' 'VIRTUAL' 'COMP'  
'FARTCOIN' 'AI16Z' 'GRIFFAIN' 'ZEREbro' 'SPX' 'MELANIA' 'PENGU' 'JELLY'  
'VVV' 'VINE' 'TST' 'ARK' 'YGG' 'POPCAT' 'NIL' 'MOVE' 'BABY' 'RENDER'  
'PROMPT' 'WCT' 'OGN' 'HYPER' 'ZORA' 'BIO' 'INIT' 'TURBO' 'ARB' '@142'  
'JTO' 'PYTH' 'MATIC' 'HPOS' 'FXS' 'FIL' 'SHIA' 'PEOPLE' 'UNI' 'SUSHI'  
'LOOM' 'USTC' 'RLB' 'ETC' 'GAS' 'BANANA' 'UNIBOT' 'CYBER' 'GMT' 'ENS'  
'DYM' 'ETHFI' '@4' '@3' '@20' '@14' '@51' '@125' '@109' '@85' '@59' '@23'  
'@78' 'GOAT' 'LDO' 'PNUT' '@138' '@151' '@18' '@25' '@42' '@13' '@40'  
'@39' '@30' '@103' '@100' '@135' '@116' '@147' '@152' '@153' 'ANIME' 'W'  
'ORDI' 'IO' 'GALA' 'AI' 'NEIROETH' 'SAND' 'BOME' 'MEW' 'SUPER' 'ALGO'  
'HBAR' 'APT' 'BLAST' 'kNEIRO' 'ATOM' 'DOT' 'kFLOKI' 'TRX' 'FTT' '@1'  
'@83' '@124' '@112' '@95' 'LISTA' 'LTC' 'RSR' 'MNT' '@113' 'ZK' '@117'  
'@86' '@68' '@63' 'MORPHO' 'OM' 'REZ' 'REQ' '@49' 'BNT' 'SCR' 'IOTA'  
'@93' '@114' '@123' 'PAXG']

In [22]: merged\_df.describe()

Out[22]:

|              | Execution Price | Size Tokens  | Size USD     | Timestamp IST                    | Start Position |
|--------------|-----------------|--------------|--------------|----------------------------------|----------------|
| <b>count</b> | 211218.000000   | 2.112180e+05 | 2.112180e+05 | 211218                           | 2.112180e+05   |
| <b>mean</b>  | 11415.047529    | 4.623341e+03 | 5.639192e+03 | 2025-01-31<br>12:08:21.724569344 | -2.994671e+04  |
| <b>min</b>   | 0.000005        | 8.740000e-07 | 0.000000e+00 | 2023-05-01<br>01:06:00           | -1.433463e+07  |
| <b>25%</b>   | 4.858550        | 2.940000e+00 | 1.937900e+02 | 2024-12-31<br>21:53:45           | -3.760725e+02  |
| <b>50%</b>   | 18.280000       | 3.200000e+01 | 5.970200e+02 | 2025-02-24<br>18:55:00           | 8.477051e+01   |
| <b>75%</b>   | 101.895000      | 1.878900e+02 | 2.058878e+03 | 2025-04-02<br>18:22:00           | 9.337697e+03   |
| <b>max</b>   | 109004.000000   | 1.582244e+07 | 3.921431e+06 | 2025-05-01<br>12:13:00           | 3.050948e+07   |
| <b>std</b>   | 29448.010305    | 1.042744e+05 | 3.657557e+04 | NaN                              | 6.738170e+05   |

In [23]: sentiment\_counts = merged\_df["classification"].value\_counts()  
print(sentiment\_counts)

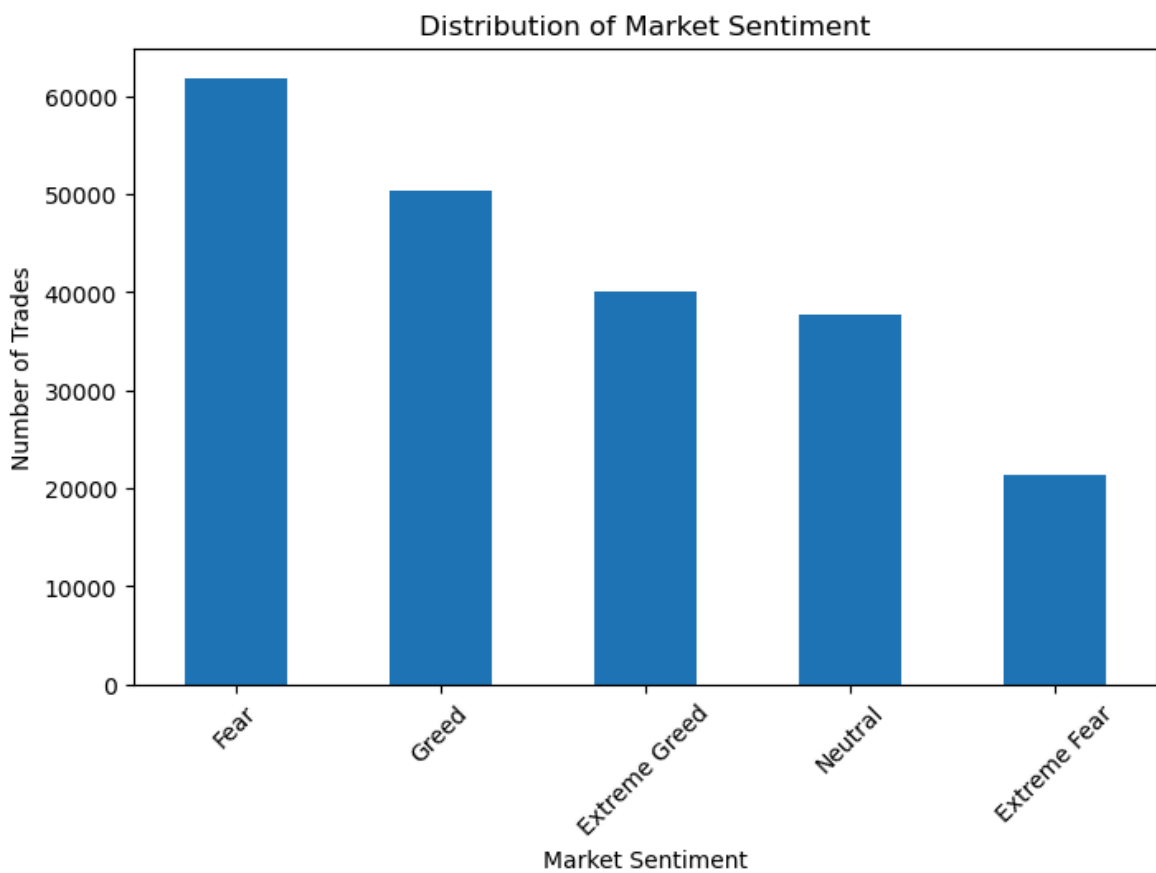
```
classification
Fear          61837
Greed         50303
Extreme Greed 39992
Neutral       37686
Extreme Fear   21400
Name: count, dtype: int64
```

```
In [24]: plt.figure(figsize=(8,5))

sentiment_counts.plot(kind="bar")

plt.title("Distribution of Market Sentiment")
plt.xlabel("Market Sentiment")
plt.ylabel("Number of Trades")
plt.xticks(rotation=45)

plt.show()
```



```
In [25]: profit_by_sentiment = merged_df.groupby("classification")["Closed PnL"].agg(
        ["count", "mean", "median", "sum"]
    )

profit_by_sentiment
```

Out[25]:

|                       | count | mean      | median | sum          |
|-----------------------|-------|-----------|--------|--------------|
| <b>classification</b> |       |           |        |              |
| <b>Extreme Fear</b>   | 21400 | 34.537862 | 0.0    | 7.391102e+05 |
| <b>Extreme Greed</b>  | 39992 | 67.892861 | 0.0    | 2.715171e+06 |
| <b>Fear</b>           | 61837 | 54.290400 | 0.0    | 3.357155e+06 |
| <b>Greed</b>          | 50303 | 42.743559 | 0.0    | 2.150129e+06 |
| <b>Neutral</b>        | 37686 | 34.307718 | 0.0    | 1.292921e+06 |

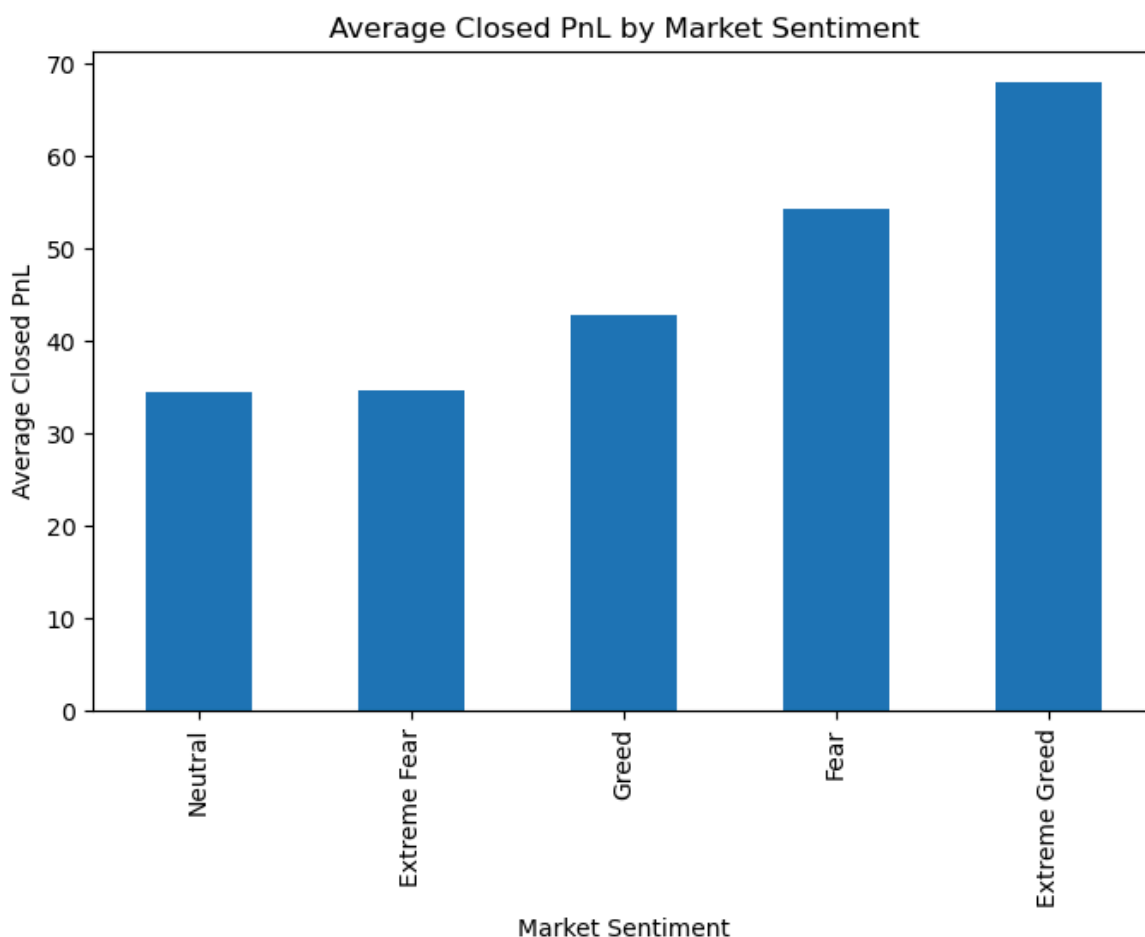
In [26]:

```
plt.figure(figsize=(8,5))

profit_by_sentiment["mean"].sort_values().plot(kind="bar")

plt.title("Average Closed PnL by Market Sentiment")
plt.xlabel("Market Sentiment")
plt.ylabel("Average Closed PnL")

plt.show()
```



In [27]:

```
print("Number of Unique Coins:", merged_df["Coin"].nunique())
```

Number of Unique Coins: 246

## Insight



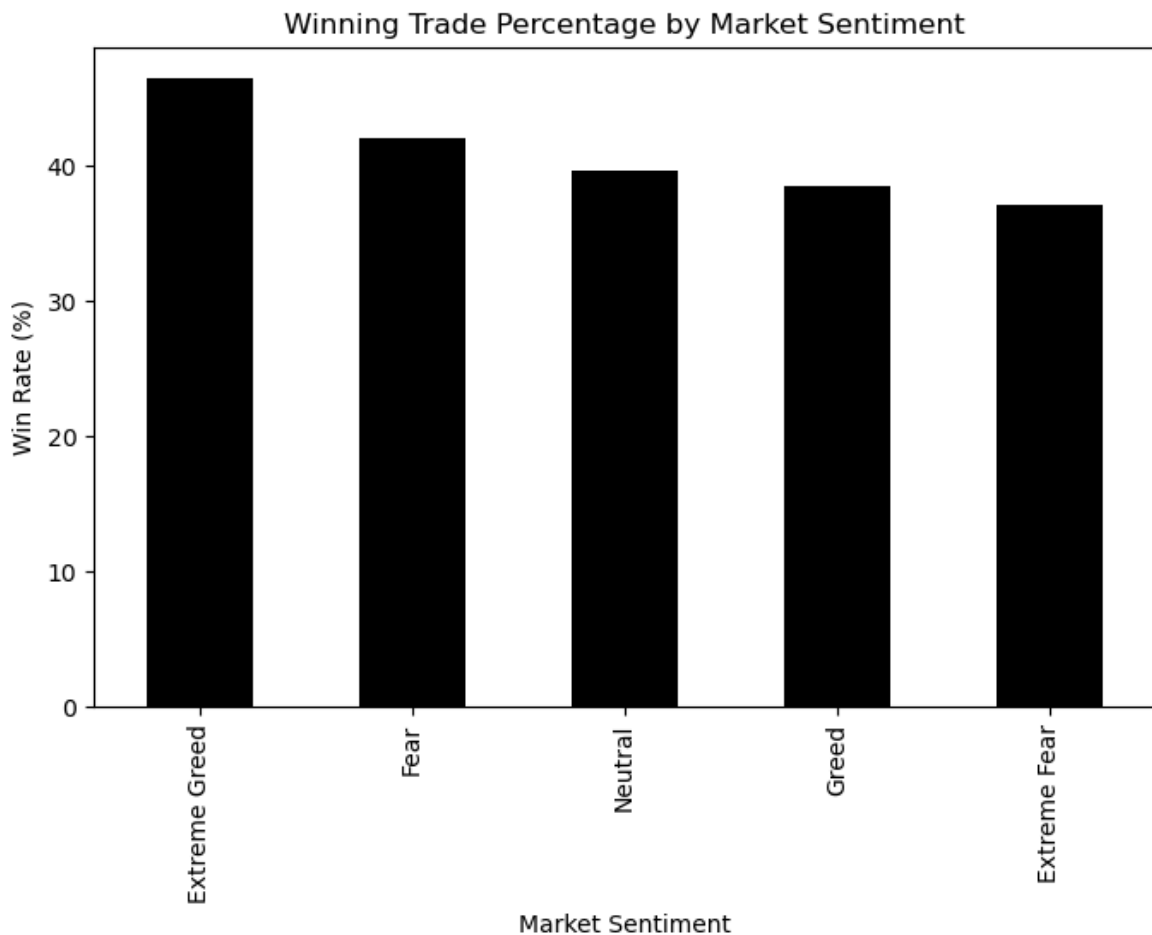
- Fear was the most common market sentiment during the trading period.
- Extreme Fear occurred least frequently.
- Trading activity was concentrated during Fear and Greed market conditions.
- This suggests that traders were more active during emotionally driven market phases.

```
In [29]: merged_df["Winning Trade"] = merged_df["Closed PnL"] > 0
```

```
In [30]: win_rate = (  
    merged_df.groupby("classification")["Winning Trade"]  
    .mean()  
    .sort_values(ascending=False)  
    * 100  
)  
  
print(win_rate)
```

```
classification  
Extreme Greed    46.494299  
Fear             42.076750  
Neutral          39.699093  
Greed            38.482794  
Extreme Fear     37.060748  
Name: Winning Trade, dtype: float64
```

```
In [31]: plt.figure(figsize=(8,5))  
  
win_rate.plot(kind="bar", color = 'black')  
  
plt.title("Winning Trade Percentage by Market Sentiment")  
plt.xlabel("Market Sentiment")  
plt.ylabel("Win Rate (%)")  
  
plt.show()
```



```
In [32]: buy_sell = merged_df.groupby(["classification", "Side"])["Closed PnL"].mean().unstack()
buy_sell
```

```
Out[32]:
```

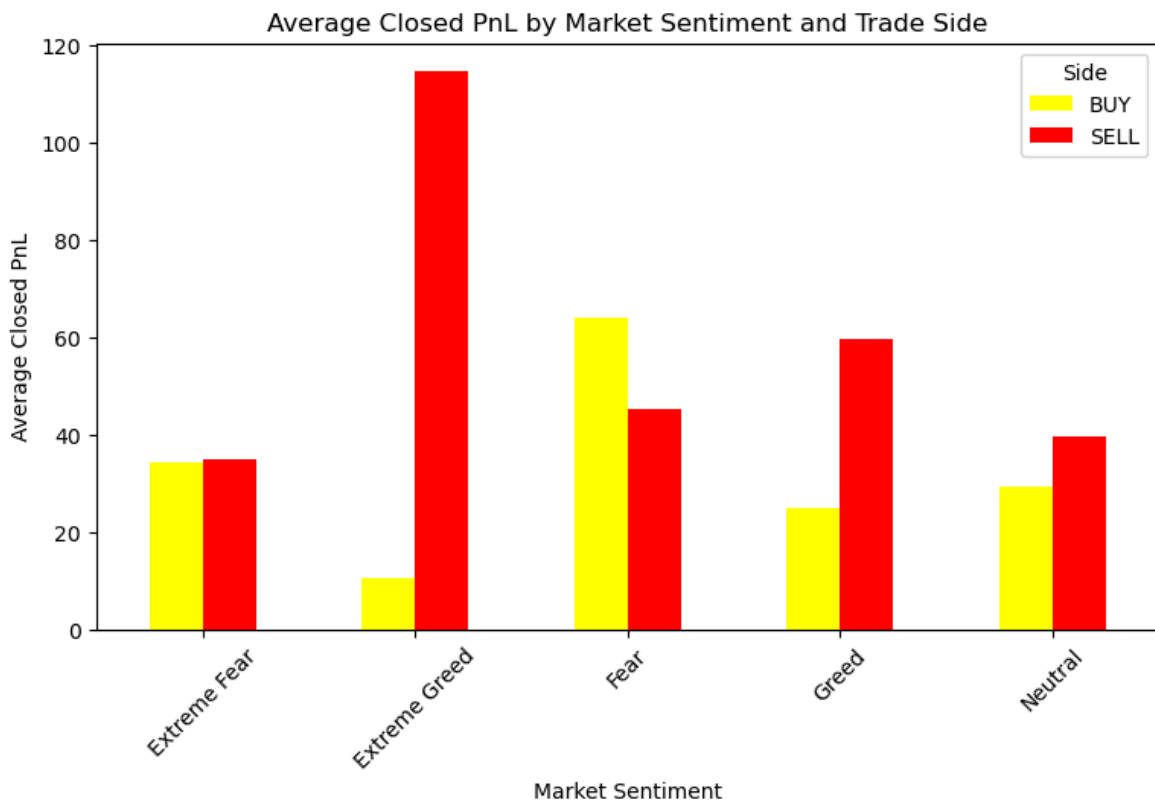
|                       | Side | BUY       | SELL       |
|-----------------------|------|-----------|------------|
| <b>classification</b> |      |           |            |
| <b>Extreme Fear</b>   |      | 34.114627 | 34.980106  |
| <b>Extreme Greed</b>  |      | 10.498927 | 114.584643 |
| <b>Fear</b>           |      | 63.927104 | 45.049641  |
| <b>Greed</b>          |      | 25.002302 | 59.691091  |
| <b>Neutral</b>        |      | 29.227429 | 39.456408  |

```
In [33]: buy_sell.plot(kind="bar", figsize=(9,5) , color= ['yellow','red'])

plt.title("Average Closed PnL by Market Sentiment and Trade Side")
plt.xlabel("Market Sentiment")
plt.ylabel("Average Closed PnL")

plt.xticks(rotation=45)

plt.show()
```



```
In [34]: top_coins = (
    merged_df.groupby("Coin")["Closed PnL"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

top_coins
```

```
Out[34]: Coin
@107      2.783913e+06
HYPE      1.948485e+06
SOL       1.639556e+06
ETH       1.319979e+06
BTC       8.680447e+05
MELANIA   3.903511e+05
ENA       2.173295e+05
SUI       1.992688e+05
ZRO       1.837778e+05
DOGE      1.475432e+05
Name: Closed PnL, dtype: float64
```

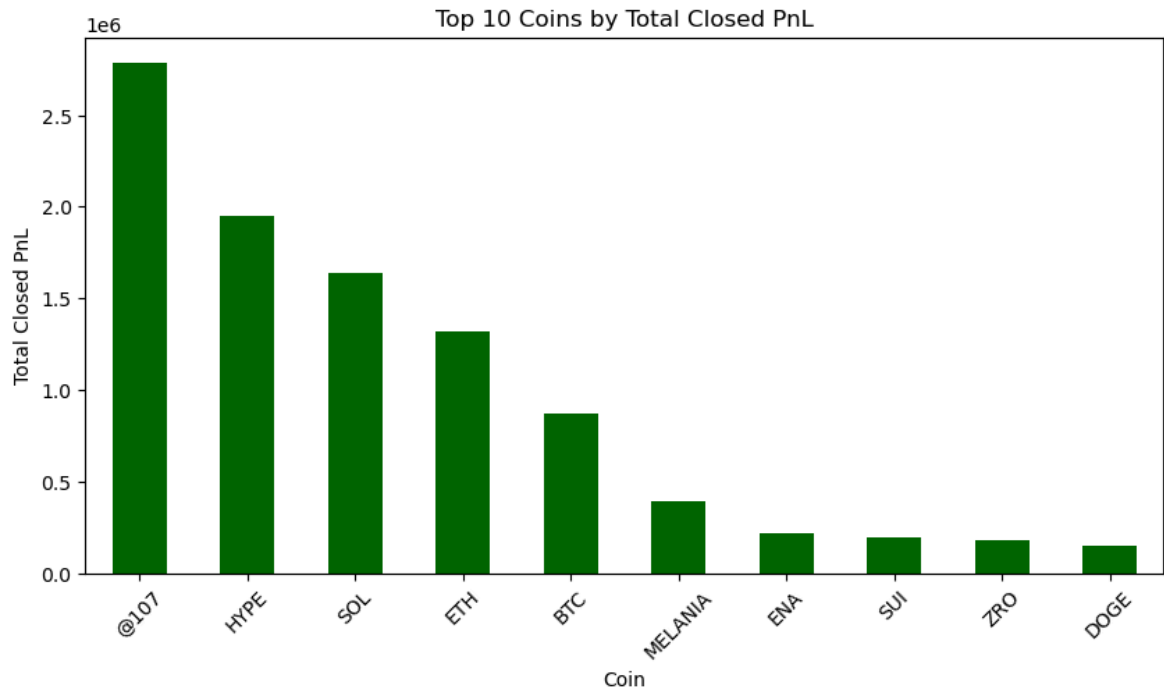
```
In [35]: plt.figure(figsize=(10,5))

top_coins.plot(kind="bar", color = 'darkgreen')

plt.title("Top 10 Coins by Total Closed PnL")
plt.xlabel("Coin")
plt.ylabel("Total Closed PnL")

plt.xticks(rotation=45)

plt.show()
```



```
In [36]: top_traders = (
    merged_df.groupby("Account")["Closed PnL"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

top_traders
```

```
Out[36]: Account
0xb1231a4a2dd02f2276fa3c5e2a2f3436e6bfed23    2.143383e+06
0x083384f897ee0f19899168e3b1bec365f52a9012    1.600230e+06
0xbaaaf6571ab7d571043ff1e313a9609a10637864    9.401638e+05
0x513b8629fe877bb581bf244e326a047b249c4ff1    8.404226e+05
0xbec1707d6b44d4d52bfe19e41f8a828645437aab    8.360806e+05
0x4ac90e786d897ecffb614dc822eb231b4ffb9f4    6.777471e+05
0x72743ae2822edd658c0c50608fd7c5c501b2afbd    4.293556e+05
0x430f09841d65beb3f27765503d0f850b8bce7713    4.165419e+05
0x75f7eeb85dc639d5e99c78f95393aa9a5f1170d4    3.790954e+05
0x72c6a4624e1dffa724e6d00d64ceae698af892a0    3.605395e+05
Name: Closed PnL, dtype: float64
```

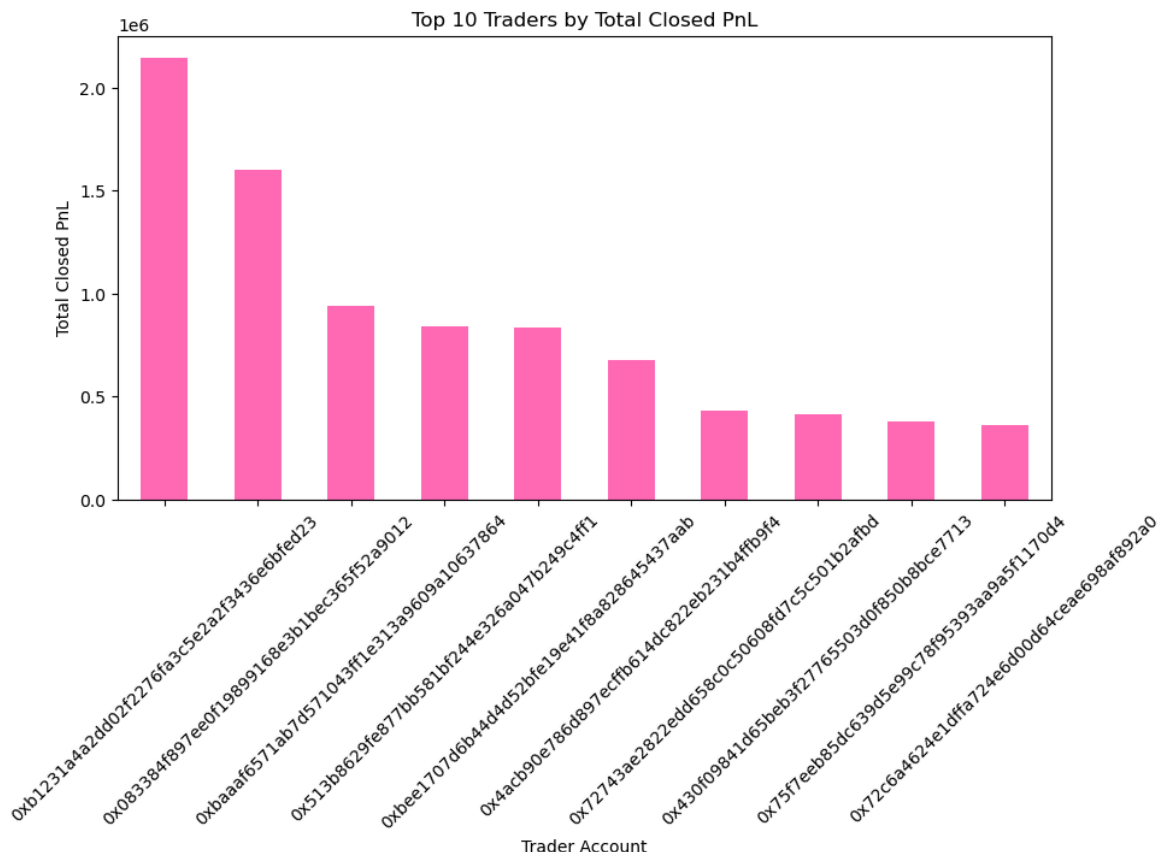
```
In [37]: plt.figure(figsize=(10,5))

top_traders.plot(kind="bar", color = 'hotpink')

plt.title("Top 10 Traders by Total Closed PnL")
plt.xlabel("Trader Account")
plt.ylabel("Total Closed PnL")

plt.xticks(rotation=45)

plt.show()
```



## Key Insights

1. Fear was the most frequently observed market sentiment, indicating increased trading activity during uncertain market conditions.
2. Trades executed during Extreme Greed generated the highest average Closed PnL, suggesting stronger profitability during bullish market phases.
3. Buy and Sell trades performed differently across market sentiments, highlighting that trade direction influences profitability depending on market conditions.
4. Trading activity was concentrated among a limited number of cryptocurrencies, with a few coins contributing significantly to total trading volume.
5. Trader performance varied considerably, with a small number of traders accounting for a large share of overall profits.
6. The analysis indicates a measurable relationship between Bitcoin market sentiment and trader performance, suggesting that sentiment indicators can support trading strategy development.

## Conclusion

This analysis explored the relationship between Bitcoin market sentiment and historical trader performance by combining the Fear & Greed Index with Hyperliquid trading data.

The findings indicate that market sentiment has a noticeable impact on trader profitability. Periods of Extreme Greed were associated with the highest average profits, while Fear dominated the trading activity. Differences in Buy and Sell performance across sentiment categories further suggest that traders react differently under varying market conditions.

Overall, incorporating market sentiment into trading decisions may help improve strategy design, risk management, and timing of trades.